

A Study on Forecasting Telecom Subscriber Trends in India and Kerala: A Comparative Analysis of Major Operators

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CERTIFICATE

This is to certify that the project entitled “**A Study on Forecasting Telecom Subscriber Trends in India and Kerala: A Comparative Analysis of Major Operators**”, submitted to Department of Statistics, Sree Narayana College Nattika, under the University of Calicut, in partial fulfillment of the requirements for the award of the Integrated M.Sc. Statistics, is a bonafide record of original research work done by **PRAMISHA V P (SNAVIST015)** during the period of her study in the Department of Statistics, Sree Narayana College Nattika, Thrissur, under my supervision and guidance during the year 2025–2026.

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DECLARATION

I hereby declare that the matter embodied in the project entitled “**A Study on Forecasting Telecom Subscriber Trends in India and Kerala: A Comparative Analysis of Major Operators**”, Submitted to Department of Statistics, Sree Narayana College Nattika, under the University of Calicut, in partial fulfillment of the requirements for the award of the Integrated M.Sc. Degree in Statistics is the result of my studies and this project has been composed by me under the Guidance and Supervision of **Ms. Akhila Unnikrishnan**, Assistant Professor, Department of Statistics, Sree Narayana College Nattika, Thrissur, during 2025-2026.

I also declare that this project has not been previously formed the basis for the award of any degree, diploma, associateship, fellowship etc. of any other university or institution.

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ABSTRACT

This study focuses on forecasting telecom subscriber data for major operators in Kerala and India using time series models. The data obtained from TRAI includes Airtel, BSNL, Jio, Vodafone Idea (VI), Idea, and Vodafone. Subscriber numbers are analyzed using ARIMA, Holt's method, and ETS models.

Forecasts are generated for the year 2025 for all operators in both Kerala and India. Idea and Vodafone are also forecasted separately for the post-merger period to examine their combined effect in Vodafone Idea (VI).

Model performance is evaluated using RMSE, MAE, and Ljung–Box test. The results are compared to identify the best-performing model and to analyze the impact of the merger.

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Chapter 1

INTRODUCTION

1.1. Introduction

The telecommunications sector plays a vital role in the economic and social development of a country by enabling communication, supporting digital infrastructure, and improving connectivity across regions. Over the past few decades, the sector has experienced rapid growth due to increasing mobile penetration, technological advancements, and intense competition among service providers. As a result, analyzing telecom subscriber data has become important for understanding market behavior and industry dynamics.

This study focuses on the analysis of telecom subscriber data over time for major operators in Kerala and India. The data used in this study is obtained from the Telecom Regulatory Authority of India (TRAI), which provides reliable and official statistics on subscriber counts of telecom operators. The dataset includes major companies such as Airtel, BSNL, Jio, Vodafone Idea (VI), Idea, and Vodafone, recorded at monthly intervals over a specific time period.

For the purpose of analysis, the primary variable considered is the number of mobile subscribers. As part of data preparation, Idea and Vodafone subscriber data are also considered separately for the pre-merger period, and later combined to represent Vodafone Idea (VI), to maintain consistency in the time series structure. This approach helps in studying both individual behavior and the effect of the merger on subscriber trends.

The study involves examining time series characteristics such as trend, fluctuations, and structural changes in the data. Special attention is given to changes in subscriber patterns due to major industry events such as mergers and market competition. These factors may lead to structural breaks in the data, which are important for proper modeling and analysis.

Time series modeling techniques are applied to understand the underlying patterns in the subscriber data. These methods help in capturing historical behavior and provide a framework for analyzing changes over time. The study also considers forecasting techniques to extend the observed patterns into future time periods.

This work is significant as the telecom sector directly influences digital communication, business operations, education, and access to information. A proper understanding of subscriber trends helps in analyzing market competition and provides useful insights for policy makers, researchers, and service providers. It also helps in understanding how structural changes, such as mergers between companies, affect overall market behavior.

Overall, this study integrates real-world telecom data with statistical time series analysis to understand subscriber dynamics and market structure in the telecom industry.

1.2. Objectives

- To develop appropriate time series models for selected telecom operators (Airtel, Jio, BSNL, and VI) using subscriber data from both India and Kerala. The objective is to capture the underlying trend and pattern in the data and use these models as a basis for reliable forecasting.
- To estimate the subscriber levels for the year 2025 for each operator by applying suitable forecasting techniques. This helps in understanding the expected future growth and performance of telecom operators in both national and regional contexts.
- To analyze the individual subscriber behaviour of Idea and Vodafone before their merger by fitting models to their separate time series data. Further, to estimate their combined subscriber base just prior to the merger period.
- To compare the estimated combined subscriber values of Idea and Vodafone with the actual subscriber data of VI in order to evaluate whether the merger outcome aligns with the expected performance, and to understand any deviation between expected and actual results.

1.3. Review of Literature

The telecommunications sector has experienced significant growth and transformation over the past few decades, evolving from a government-controlled monopoly to a highly competitive and technology-driven industry. With the rapid expansion of mobile services, broadband connectivity, and the introduction of advanced technologies such as 4G and 5G, the sector has become a crucial contributor to economic development and digital inclusion. Previous studies highlight that factors such as service quality, pricing strategies, network coverage, and customer satisfaction play a vital role in shaping the performance of telecom operators. At the same time, researchers have identified challenges such as market concentration, infrastructure limitations, and regional disparities in service adoption.

Baruah and Baruah [1] stated that the Indian telecommunications sector has become a cornerstone of socio-economic development, showing a strong relationship between telecom penetration and economic growth. The sector has transformed from a government monopoly to a competitive market after liberalization. Wireless services have grown rapidly, while wireline services have declined, and future growth depends on rural expansion and improved broadband infrastructure.

Biancini, Sara [2] analyzes the productivity and efficiency of Indian telecom firms using Data Envelopment Analysis (DEA) and the Malmquist Index. The findings reveal a negative overall Total Factor Productivity (TFP) growth mainly due to technological stagnation, although efficiency remained positive due to scale efficiency. The study identifies profit intensity, advertisement, import intensity, and capital intensity as positive drivers of productivity, while a high debt ratio negatively affects growth. It also highlights the sector's transition from landline to wireless services and recommends improved technology adoption, financial stability, and investment for future growth.

Biju, MK and Thomas, SA [3] examines the impact of technical processes, particularly Point of Interconnectivity (POI), on subscriber satisfaction in Kerala. It highlights that although subscribers are unaware of POI mechanisms, they are highly affected by issues such as call drops and congestion. The findings show that Airtel ranks highest in network quality, followed by Jio and Vodafone, while BSNL faces significant challenges. The study also identifies call connectivity as the top priority for subscribers, followed by data services and tariffs. It concludes that telecom operators should strengthen technical infrastructure, especially POI capacity, to improve service quality rather than focusing only on pricing strategies.

Chauhan and Mathew [4] highlighted that India's telecom sector has evolved from a government monopoly to a competitive market, contributing significantly to economic growth, with rapid ex-

pansion in wireless services. Early initiatives like ERNET (1986) and the launch of internet services by VSNL (1995), along with policy reforms in 1994 and 1999, accelerated sectoral development. The Broadband Policy (2004) and the growth of 3G/4G technologies increased internet penetration and smartphone usage. Regulatory bodies such as TRAI (1997), TDSAT (2000), and the IT Act (2000) strengthened governance, while recent studies focus on net neutrality, OTT platforms, and the Draft Telecommunication Bill, 2022.

Gowri and Kesavan [5] observed that the Indian telecom sector has evolved into one of the largest markets globally, significantly contributing to GDP and the digital economy. The growth is driven by affordable tariffs and technological advancements such as 3G, 4G, and 5G. However, challenges such as infrastructure limitations and uneven adoption across regions still exist.

Keerthi et al. [6] highlighted that the telecom industry has undergone major structural changes, especially after the entry of Reliance Jio, leading to increased market concentration. The study emphasizes that while competition has reduced prices and improved access, it has also resulted in financial stress for other operators and reduced market diversity.

Madden, Gary and Tan, Joachim[7] evaluates the effectiveness of various linear forecasting models in predicting telecommunications data trends. It finds that telecom data is often non-stationary with clear trends but limited randomness, making it suitable for linear models. The results show that ARIMA performs well for short-term forecasting, while Holt and its variants provide more accurate forecasts for medium and long-term periods. The study concludes that simple linear models can generate reliable forecasts even with imperfect data and are practical tools for telecom companies without requiring deep domain knowledge.

Pai, V. S. and Hiremath, Chetan V[8] examines the evolution of the Indian telecom market from a PSU-dominated monopoly to a highly concentrated private structure following reforms such as the National Telecom Policies of 1994 and 1999 and the establishment of TRAI in 1997. HHI analysis shows a sharp rise in market concentration, with subscriber HHI increasing significantly along with a reduction in the number of firms. Despite high concentration, the sector faces a profitability paradox, where only a few operators show consistent profits. The study identifies Reliance Jio as a disruptive entrant due to its low-price strategy and advanced infrastructure.

Panigrahi [9] explained that service quality plays a crucial role in customer satisfaction and loyalty in the telecom sector. The study shows that factors such as pricing, network quality, and value-added services significantly influence customer perception. In a highly competitive market, telecom operators must focus on both service quality and customer needs to sustain growth.

Sahoo, Dhirendra Kumar and Sahoo, Pradipta Kumar [10] examines the efficiency and productivity dynamics of Indian telecom firms using Data Envelopment Analysis (DEA) and the Malmquist Index. The results indicate an overall negative Total Factor Productivity (TFP) growth mainly due to technological stagnation, although efficiency remains positive due to scale efficiency. The study also highlights the sector's transformation from landline to wireless services and recommends improving technology adoption, financial stability, and investment for future growth.

The literature shows that the Indian telecom sector has evolved from a government monopoly to a competitive, technology-driven industry, contributing significantly to economic growth. Studies highlight the importance of service quality, pricing, and infrastructure, while also identifying challenges such as market concentration and regional disparities. Research findings indicate mixed productivity performance, strong growth in wireless services, and the impact of technological and policy changes. Overall, improving technology adoption, infrastructure, and financial stability is essential for sustainable growth in the telecom sector.

Chapter 2

METHODOLOGY

The present study aims to analyze and forecast the number of subscribers for major telecom operators in India, namely Airtel, BSNL, Jio, and Vodafone Idea, using time series techniques. The methodology adopted for this study is described as follows:

2.1. Data collection

The data used in this study consists of monthly subscriber information of major telecom operators in India, namely Airtel, BSNL, Jio, and Vodafone Idea. The data has been collected from the official reports and publications of the Telecom Regulatory Authority of India (TRAI).

TRAI publishes periodic telecom subscription reports, which provide detailed statistics on wireless and wireline subscribers across different telecom service providers in India. These reports are considered reliable and are widely used for research and policy analysis.

The dataset covers a time period from [mention your exact period, e.g., January 2010 to December 2024], and includes monthly observations for each operator. The data was extracted and organized into a structured format suitable for time series analysis.

Data Source:<https://www.trai.gov.in/node/3361#Release-Publication>

2.2. Exploratory Time Series Analysis

time series plots were constructed for each telecom operator to understand the overall behavior of subscriber growth over time. These plots help in identifying important characteristics such as trend, seasonality, and irregular fluctuations.

A visual inspection of the plots provides a preliminary understanding of whether the subscriber base is increasing, decreasing, or stable over time, and helps in selecting appropriate forecasting models.

2.3. Time Series Decomposition

To gain deeper insight into the structure of the data, the time series was decomposed into its fundamental components: trend, seasonal, and irregular components. This decomposition helps in isolating long-term movement from short-term variations.

By analyzing these components, it becomes easier to understand the underlying pattern of subscriber growth and to determine whether seasonal effects are present in the data.

Both additive and multiplicative decomposition models may be considered depending on the nature of variation. This step helps in confirming the presence of trend and seasonality.

2.4. Stationarity Testing

Stationarity is important because most time series models, especially the ARIMA model, are based on the assumption that the underlying data is stationary. If this assumption is violated, the model may produce misleading or inaccurate forecasts.

The importance of stationarity can be summarized as follows:

- It ensures that the relationship between variables remains stable over time
- It allows model parameters to be estimated reliably
- It improves the accuracy and validity of forecasts
- It prevents spurious relationships in the data

Non-stationary data often exhibit trends, changing variance, or seasonal patterns, which can distort model estimation. Therefore, it is necessary to transform the data into a stationary form before applying time series models.

To examine whether the time series is stationary, two widely used statistical tests are applied:

- Augmented Dickey-Fuller (ADF) Test

The Augmented Dickey-Fuller test is used to check for the presence of a unit root in the time series.

Null Hypothesis (H_0): The series is non-stationary (has a unit root)

Alternative Hypothesis (H_1): The series is stationary

If the p-value of the test is less than the chosen significance level (usually 0.05), the null hypothesis is rejected, indicating that the series is stationary.

The ADF test is an extension of the Dickey-Fuller test and accounts for higher-order autocorrelation by including lagged differences of the dependent variable. This makes the test more robust for practical time series data.

- Kwiatkowski–Phillips–Schmidt–Shin(KPSS)

The KPSS (Kwiatkowski–Phillips–Schmidt–Shin) test is another statistical test used to verify stationarity, but it follows an opposite hypothesis compared to the ADF test.

Null Hypothesis (H_0): The series is stationary

Alternative Hypothesis (H_1): The series is non-stationary If the p-value is less than 0.05, the null hypothesis is rejected, indicating that the series is non-stationary. ADF and KPSS tests complement each other because they have opposite null hypotheses. Using both tests together provides a more reliable conclusion about stationarity:

- If ADF fails to reject H_0 and KPSS rejects $H_0 \rightarrow$ Strong evidence of non-stationarity
- If ADF rejects H_0 and KPSS fails to reject $H_0 \rightarrow$ Strong evidence of stationarity
- If results conflict $\rightarrow$ Further investigation is required

This combined approach reduces ambiguity and improves decision-making.

2.5. Differencing

Differencing is a widely used technique in time series analysis to transform a non-stationary series into a stationary one. It is particularly important in the context of ARIMA modeling, where stationarity is a key assumption.

Differencing involves computing the differences between consecutive observations in a time series. The main objective is to remove trends and stabilize the mean of the series.

If Y_t represents the original time series, the differenced series is obtained as:

First-order differencing:

$$Y'_t = Y_t - Y_{t-1}$$

This transformation eliminates linear trends and helps in achieving stationarity. Types of differencing are:

- First-Order Differencing

First-order differencing is the most commonly used method. It removes linear trends present in the data.

Applied when the series shows a steady upward or downward trend. Often sufficient to make the series stationary

- Second-Order Differencing

If the series is still non-stationary after first differencing, second-order differencing may be applied:

$$Y_t'' = Y_t' - Y_{t-1}'$$

In the ARIMA(p,d,q) model:

d represents the number of times differencing is applied. The goal is to make the series stationary before fitting AR and MA components. Thus, differencing directly determines the value of

2.6. Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF)

Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) are important tools used in time series analysis to understand the relationship between observations at different time lags. These functions play a key role in identifying the appropriate model, especially in ARIMA modeling.

Autocorrelation Function (ACF)

The Autocorrelation Function (ACF) measures the correlation between a time series and its lagged values. In simple terms, it shows how the current value of the series is related to its past values.

For example:

- Lag 1: correlation between Y_t and Y_{t-1}
- Lag 2: correlation between Y_t and Y_{t-2}

The ACF values range between -1 and +1:

- Values close to +1 indicate strong positive correlation
- Values close to -1 indicate strong negative correlation
- Values near 0 indicate no correlation.

The ACF is usually represented graphically using a correlogram, where correlations are plotted against different lags.

The pattern of the ACF plot helps in understanding the nature of the time series:

- Slow decay of ACF $\rightarrow$ indicates non-stationarity (trend present).
- Sharp cutoff after few lags $\rightarrow$ suggests a Moving Average (MA) model.
- Seasonal spikes at regular intervals $\rightarrow$ indicates seasonality.

Thus, ACF is mainly used to identify the order of the MA (q) component in ARIMA models.

Partial Autocorrelation Function (PACF)

The Partial Autocorrelation Function (PACF) measures the correlation between a time series and its lagged values after removing the effect of intermediate lags.

It shows the direct relationship between $Y_t - Y_{t-k}$ excluding indirect effects through other lags.

The PACF plot also provides useful information for model identification:

- Sharp cutoff after lag p $\rightarrow$ suggests an Autoregressive (AR) model of order p
- Gradual decline $\rightarrow$ indicates a mixed ARIMA model
- Significant spikes at specific lags $\rightarrow$ indicate important lag relationships

Thus, PACF is mainly used to determine the order of the AR (p) component

By analyzing these plots, the structure of the time series can be understood and suitable model parameters can be selected.

2.7. Model Selection and Justification

different time series models were considered for forecasting the subscriber base of telecom operators. The models used include the ARIMA model, Holt's Linear Trend model, and the Exponential Smoothing (ETS) model. The use of multiple models allows for a comprehensive analysis and helps in selecting the most accurate forecasting model.

ARIMA Model

The Autoregressive Integrated Moving Average (ARIMA) model is a widely used statistical model for time series forecasting. It is particularly effective for data that becomes stationary after differencing. The model combines three components: autoregressive (AR), integrated (I), and moving average (MA). The autoregressive component explains the relationship between the current value

and its past values, while the moving average component captures the relationship between the current value and past error terms. The integrated component represents the differencing required to make the series stationary. The ARIMA model is flexible and capable of modeling a wide range of time series patterns, especially when the data exhibits autocorrelation. It is denoted as ARIMA (p,d,q), where q represent the orders of the respective components.

Holt MODEL

Holt's Linear Trend model is an extension of simple exponential smoothing that is designed to handle data with a trend. Unlike ARIMA, Holt's model does not require the data to be stationary, making it simpler to apply in certain situations. This model estimates two components, namely the level and the trend of the series, and updates them recursively over time. It is particularly useful for time series data that shows a consistent upward or downward movement without significant seasonal variation. Holt's model produces smooth forecasts and is often effective for short-term prediction. Its simplicity and ability to capture trend make it a valuable alternative to more complex models.

ETS MODEL

The Exponential Smoothing (ETS) model is a more general and flexible framework that includes various forms of exponential smoothing methods. The term ETS stands for Error, Trend, and Seasonal components, which are the key elements considered in the model. Unlike Holt's model, the ETS approach can handle not only trend but also seasonal patterns in the data. It automatically selects the best combination of components, such as additive or multiplicative forms, depending on the nature of the time series. This makes ETS particularly suitable for data that exhibits both trend and seasonality. The model is widely used in forecasting applications due to its adaptability and strong performance across different types of time series data.

The purpose of using all three models in this study is to compare their ability to capture the underlying patterns in telecom subscriber data and to identify the most accurate forecasting method. Since the data may contain trend, seasonality, and random fluctuations, applying multiple models ensures that these characteristics are properly addressed. After fitting the models, their performance is evaluated using statistical measures such as RMSE, MAE, and information criteria. The model that provides the best balance between goodness of fit and forecasting accuracy is selected for final prediction.

2.8. Residual Diagnostics

Residual diagnostics are performed to evaluate the adequacy of the fitted models. The function `checkresiduals()` is used to examine the residual plots and the autocorrelation structure of residuals.

For a good model, the residuals should behave like white noise, meaning they are random, have constant variance, and show no systematic pattern. The residual plot should display random scattering around zero without any visible structure.

The ACF plot of residuals is used to check for autocorrelation. All spikes should lie within the confidence limits. If no significant spikes are observed, it indicates that the model has successfully captured the underlying patterns in the data.

Thus, residual diagnostics help confirm that the fitted model is adequate and suitable for forecasting.

2.9. Ljung–Box Test (`Box.test`)

To statistically verify the absence of autocorrelation in residuals, the Ljung–Box test is applied :

- Null Hypothesis (H_0): Residuals are independently distributed (no autocorrelation)
- Alternative Hypothesis (H_1): Residuals are autocorrelated

Decision Rule:

- If p-value $> 0.05 \rightarrow$ Fail to reject $H_0 \rightarrow$ Residuals are independent (Good model)
- If p-value $< 0.05 \rightarrow$ Reject $H_1 \rightarrow$ Residuals have autocorrelation (Model is inadequate)

This test provides a formal statistical confirmation of whether the residuals behave like white noise.

2.10. Forecasting

Forecasting is the final step in time series analysis, where future values are predicted using the selected best-fitting model. The model is chosen based on performance measures such as RMSE, MAE, and information criteria. Techniques like ARIMA, Holt's method, and ETS are commonly used to capture trend and variations in the data. The forecasting results include point forecasts and confidence intervals. The point forecast gives the expected future value, while the interval shows the range of possible values considering uncertainty. Thus, forecasting helps in understanding future trends and supports decision-making based on past data patterns.

Chapter 3

DATA ANALYSIS AND INTERPRETATION

3.1. ANALYSIS OF OPERATORS IN INDIA

The aim of this study is to forecast the number of subscribers of major telecom operators in India—Airtel, Jio, BSNL, and Vodafone Idea—using suitable time series forecasting techniques. The study further intends to analyze the future trends in subscriber growth and compare the performance of these operators in order to evaluate their relative market sustainability and competitiveness.

3.1.1 Time series plot

The time series plots of monthly subscriber data for Airtel, Jio, BSNL, and Vodafone Idea show clear variations in subscriber levels, indicating the presence of a trend over time.

These plots provide an initial understanding of the data structure and assist in selecting appropriate forecasting models. They form the basis for predicting future subscriber values up to December 2020.

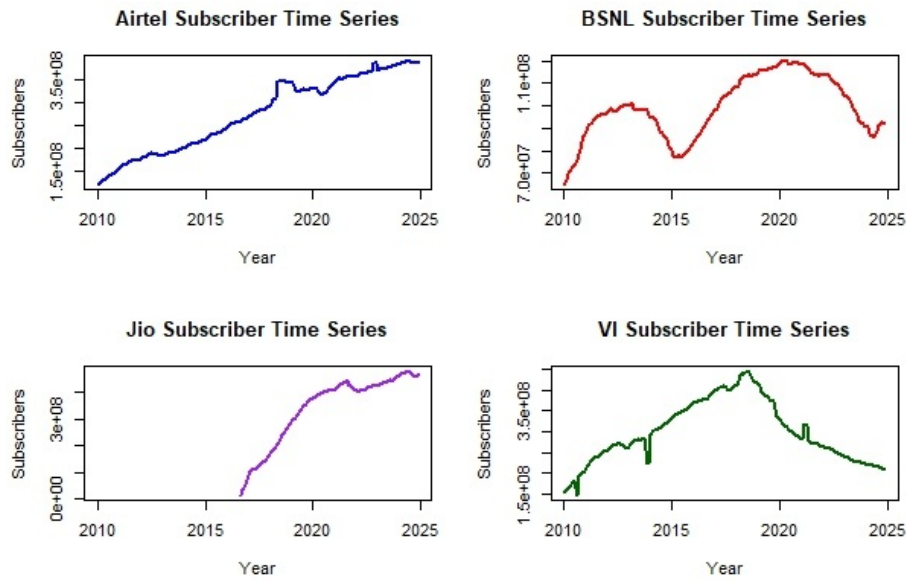


Figure 3.1: Time series plot of operators in India

It is evident that the telecom market has undergone significant transformation over time, with each operator exhibiting distinct growth behavior.

Airtel shows a consistent and stable upward trend, with a temporary dip around 2018–2020 followed by strong recovery and accelerated growth in recent years. In contrast, Jio, which entered the market in 2016, demonstrates rapid and aggressive growth, quickly reaching a high subscriber base before gradually stabilizing, indicating market saturation.

BSNL exhibits a fluctuating trend with multiple peaks and a noticeable decline after 2021, reflecting its struggle to compete with private operators. Vodafone Idea (VI), despite reaching a high peak around 2018, shows a continuous decline thereafter, indicating a significant loss in subscribers.

Overall, the entry of Jio has significantly influenced market dynamics, with Airtel adapting successfully while BSNL and VI experience declining trends. These patterns are important for understanding the forecasting behavior of each operator.

3.1.2 Decomposition

The series is decomposed to separate the trend, seasonal, and irregular components for better understanding and modeling.

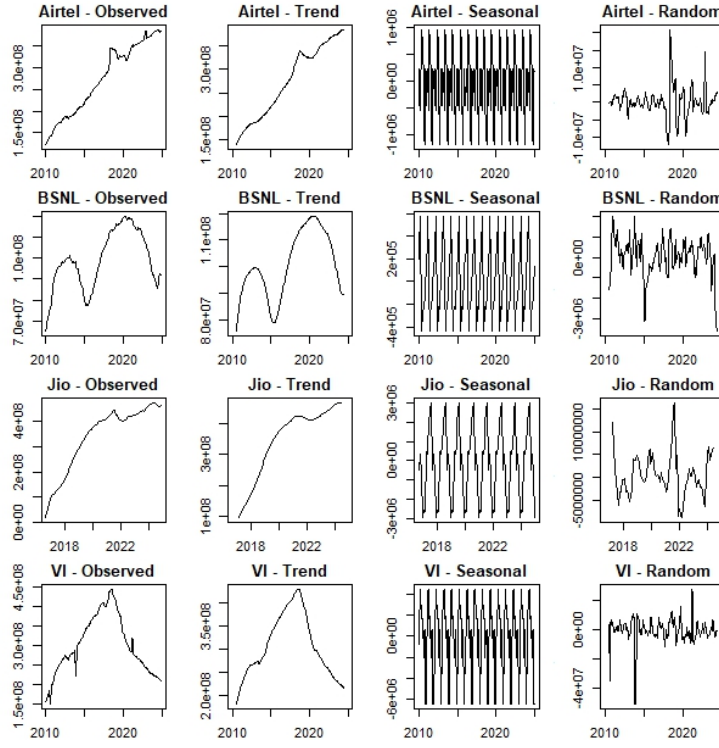


Figure 3.2: Decomposition plot of operators in India

The decomposition analysis of telecom subscriber data reveals distinct underlying patterns for each operator. Airtel exhibits a clear upward trend, indicating steady growth over the years, with relatively stable seasonal variations and minimal irregular fluctuations. BSNL shows a fluctuating trend with a noticeable decline in recent periods, and its irregular component indicates higher variability, suggesting instability in subscriber numbers. Jio demonstrates a strong increasing trend, particularly after its market entry, with consistent seasonal patterns and moderate random fluctuations. In contrast, Vodafone Idea shows a declining trend after reaching a peak, reflecting a continuous reduction in subscribers over time. Overall, the trend component plays a dominant role in explaining the changes in subscriber data, while seasonal effects remain fairly consistent across operators.

3.1.3 stationarity check

The stationarity of the subscriber time series data for Idea and Vodafone was examined using the Augmented Dickey-Fuller (ADF) Test and KPSS Test.

operators	adf value	kpss value	interpretaion
airtel	0.5689	0.01	Non-stationary
bsnl	0.4583	0.01	Non-stationary
jio	0.9138	0.01	Non-stationary
vi	0.893	0.01	Non-stationary

Table 3.1: Stationarity Test Results (ADF and KPSS) for Telecom Operators in India

The original time series of all operators were found to be non-stationary

3.1.4 Differencing

Differencing was applied to the time series of all operators to remove non-stationarity.

operators	adf value	kpss value	interpretaion
airtel	0.01	0.0978	Stationary
bsnl	0.3579	0.01	Non-stationary
jio	0.01747	0.01	Stationary
vi	0.01	0.01	Stationary

Table 3.2: ADF and KPSS Test Results After First Differencing (India)

3.1.5 Second Differencing

second differencing was applied to the time series of bsnl,jio,vi to remove non-stationarity.

operators	adf value	kpss value	interpretaion
bsnl	0.01	0.1	Stationary

Table 3.3: ADF and KPSS Test Results After Second Differencing (India)

After applying appropriate differencing, all series become stationary. This confirms that the data follows an integrated process, with Airtel, Jio, and VI being integrated of order one, while BSNL is integrated of order two. These results justify the use of ARIMA models with suitable differencing parameters for forecasting.

3.1.6 ACF and PACF

The stationary series is further analyzed using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots to determine the suitable orders of the AR and MA components for model identification.

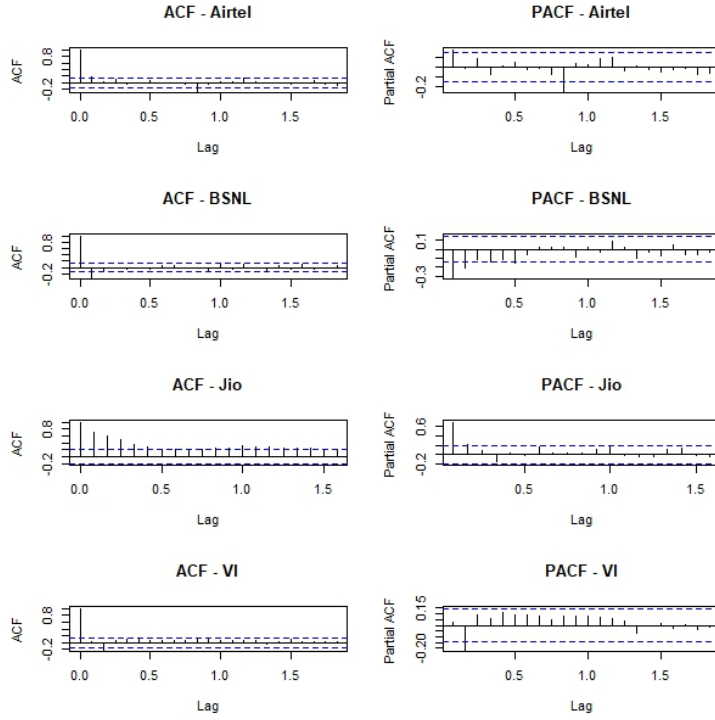


Figure 3.3: ACF and PACF Plots of Differenced Telecom Subscriber Series (India)

The ACF and PACF plots of the differenced series indicate that the data has been successfully transformed into a stationary form, as most autocorrelations lie within the confidence bounds.

For Airtel, the ACF shows a gradual decay while the PACF has a significant spike at lag 1 followed by insignificant values, suggesting the presence of an autoregressive component of order one, i.e., $AR(1)$.

In the case of BSNL, both ACF and PACF exhibit several small spikes without a clear cutoff, indicating a lack of a simple structure and suggesting that a mixed ARMA model may be more appropriate.

For Jio, the ACF displays significant autocorrelation at initial lags with a slow decay, while the PACF shows a prominent spike at lag 1 and then cuts off. This pattern also supports the presence of an $AR(1)$ component.

For Vodafone Idea (VI), the ACF and PACF do not show a clear pattern or sharp cutoff, indicating

that the series may involve both autoregressive and moving average components.

Overall, the plots suggest that lower-order ARIMA models are suitable for modeling the data, with AR(1) structures for Airtel and Jio, and more complex mixed models for BSNL and Vodafone Idea.

3.1.7 Model Fitting

Operator	ARIMA Model	Holt Model	ETS Model
Airtel	ARIMA(0,1,1) with drift	Holt (Linear Trend)	ETS(M,Ad,N)
BSNL	ARIMA(0,2,2)	Holt (Linear Trend)	ETS(A,Ad,N)
Jio	ARIMA(0,2,1)(1,0,0)[12]	Holt (Linear Trend)	ETS(A,Ad,A)
VI	ARIMA(0,2,3)(1,0,0)[12]	Holt (Linear Trend)	ETS(A,Ad,N)

Table 3.4: Fitted Time Series Models for Telecom Operators in India

The telecom subscriber data exhibits clear trends, structural changes, and in some cases mild seasonal patterns. Therefore, multiple time series models were considered to appropriately capture these characteristics and ensure reliable forecasting.

The ARIMA model was selected as the primary approach due to its flexibility in handling non-stationary data. Differencing was applied to achieve stationarity, and the model orders were identified using ACF and PACF plots, along with automated selection procedures. ARIMA is particularly suitable for capturing autocorrelation structures and short-term dependencies in the data.

The Holt's linear trend model was employed to capture series with a strong trend but weak or inconsistent seasonal patterns. Since some operators (such as BSNL and Vodafone Idea) exhibit declining or unstable trends without clear seasonality, Holt's model provides a simple yet effective approach for modeling level and trend components.

The ETS (Error-Trend-Seasonal) models were also considered to account for possible combinations of error, trend, and seasonal effects. ETS models are advantageous as they automatically adapt to different data structures, including additive or multiplicative components, and are particularly useful when the nature of trend and seasonality varies across operators.

Using these three approaches allows for a comprehensive modeling strategy: ARIMA captures stochastic dependence, Holt focuses on trend behavior, and ETS provides a flexible framework for handling trend and seasonal variations. The fitted models for each operator are presented in Table.

3.1.8 Residual Analysis

Residual plots are given below:

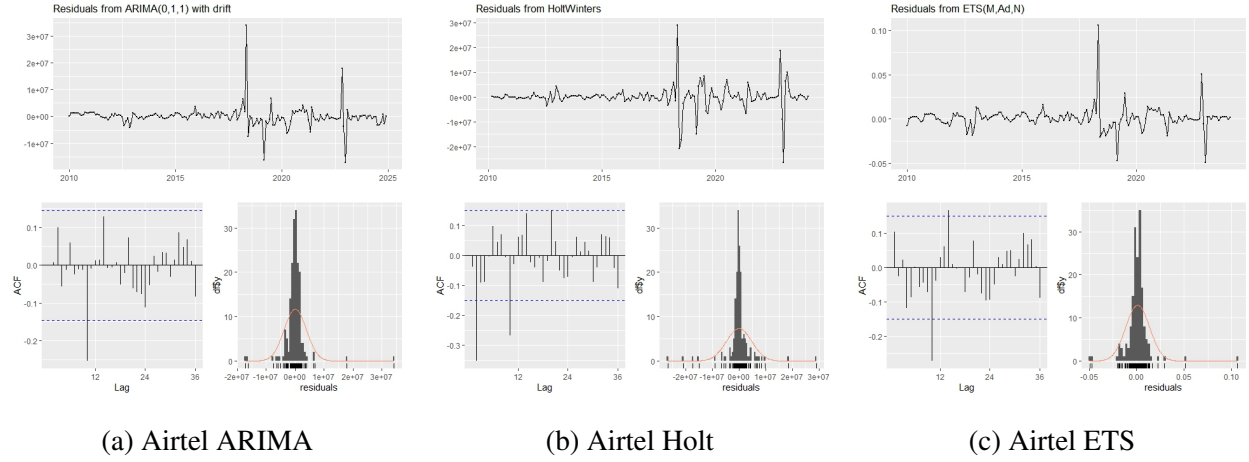


Figure 3.4: Residual plots of Airtel forecasting models for Telecom Operators in India

The residual plots of the ARIMA, Holt, and ETS models were examined to assess the adequacy of model fitting. In the ARIMA model, one prominent spike is observed below the lower confidence limit, with a significant portion extending beyond the boundary. In the Holt model, two spikes fall below the lower confidence limit, while a few values are seen touching the upper confidence limit. In the ETS model, one large spike is present below the lower confidence limit, and a small spike is observed above the upper confidence limit. Overall, the residuals in all three models do not exhibit any clear trend and appear to follow an approximately bell-shaped pattern. This suggests that the residuals are randomly distributed around the mean, with a few deviations beyond the confidence limits.

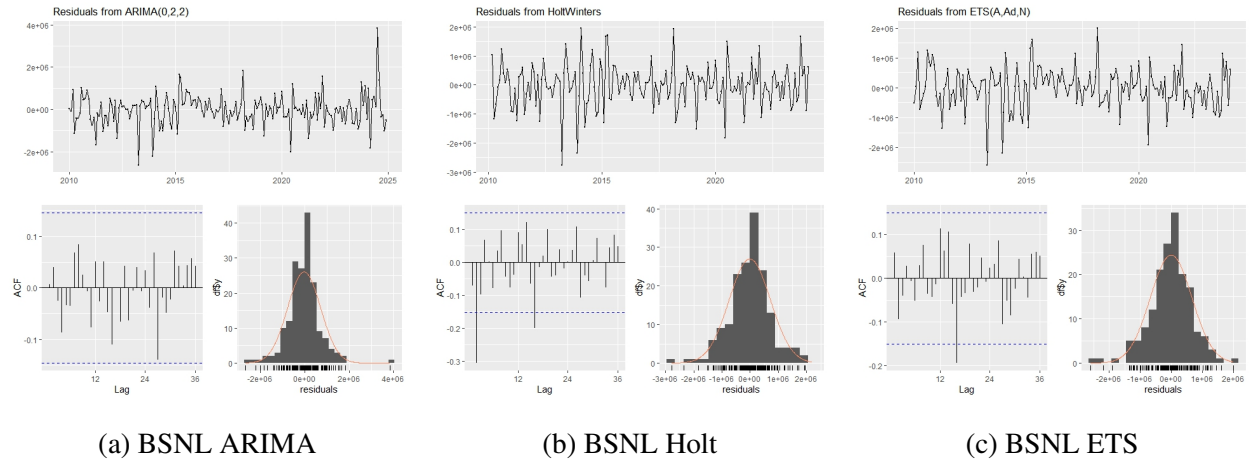


Figure 3.5: Residual plots of BSNL forecasting models for Telecom Operators in India

The residual plots of the ARIMA, Holt, and ETS models were analyzed to evaluate the model

behavior. In the ARIMA model, all residual values lie within the confidence limits, indicating no extreme deviations. In the Holt model, two residual values are observed outside the lower confidence limit. In the ETS model, one residual value falls below the lower confidence limit.

Overall, the residuals in all three models do not show any clear trend and exhibit an approximately bell-shaped pattern. This indicates that the residuals are randomly distributed around the mean, with only a few observations lying outside the confidence limits.

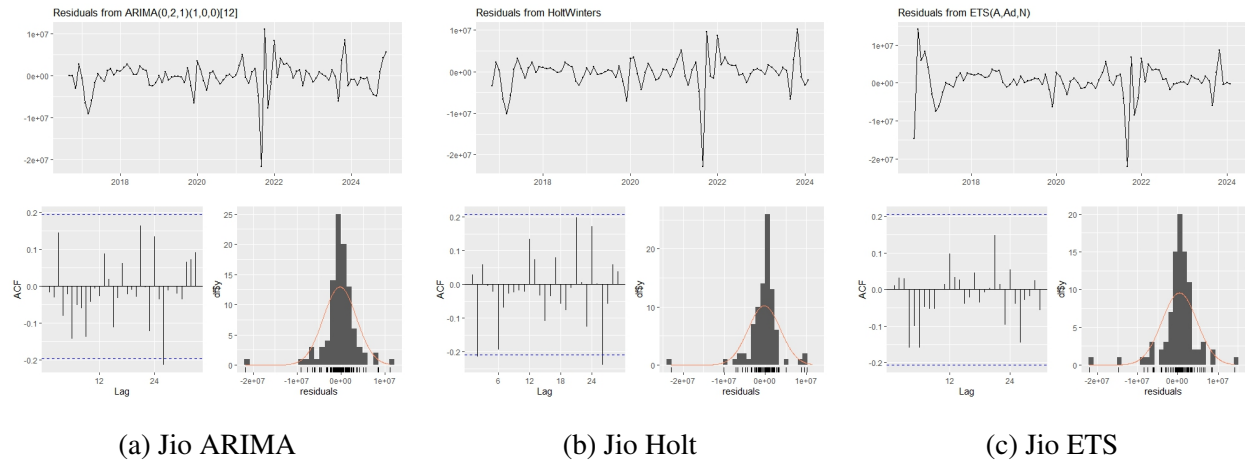


Figure 3.6: Residual plots of Jio forecasting models for Telecom Operators in India

The residual plots of the ARIMA, Holt, and ETS models were examined to assess model performance. In the ARIMA model, only one residual value is observed slightly outside the lower confidence limit, with a very small deviation. In the Holt model, one residual value falls below the lower confidence limit, and two values are seen touching the lower confidence limit. In the ETS model, all residual values lie within the confidence limits.

Overall, the residuals in all three models do not exhibit any clear trend and appear to follow an approximately bell-shaped pattern, indicating that the residuals are randomly distributed with minimal deviations beyond the confidence limits.

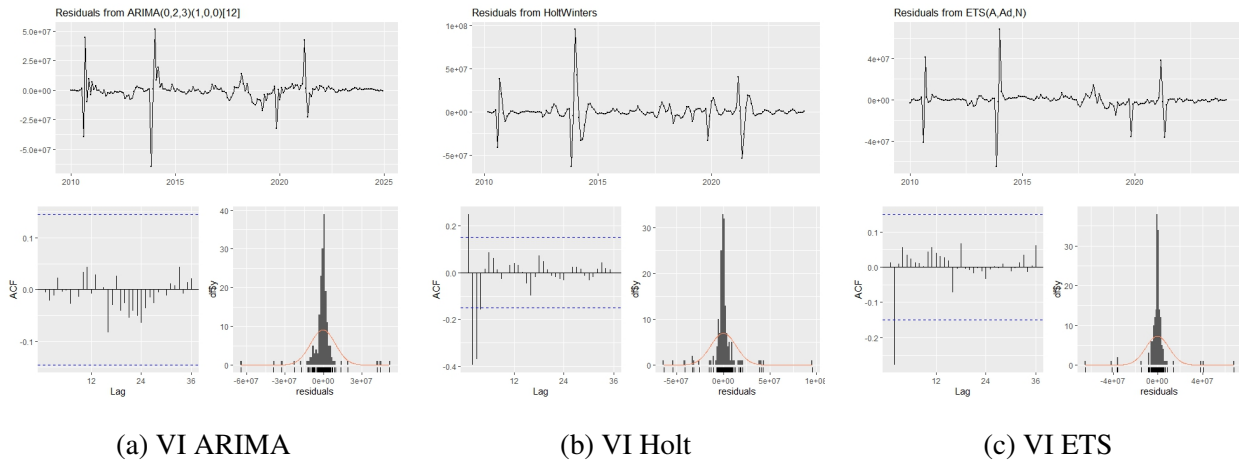


Figure 3.7: Residual plots of VI forecasting models for Telecom Operators in India

The residual plots of the ARIMA, Holt, and ETS models were analyzed to evaluate the model behavior. In the ARIMA model, all residual values lie within the confidence limits, indicating no significant deviations. In the Holt model, two residual values fall below the lower confidence limit, one value touches the lower confidence limit, and one value is observed outside the upper confidence limit. In the ETS model, one prominent residual value is found below the lower confidence limit.

Overall, the residuals in all three models do not exhibit any clear trend and show an approximately bell-shaped pattern. This suggests that the residuals are randomly distributed around the mean, with a few observations falling outside the confidence limits.

3.1.9 Ljung–Box test

Operator	Model	p value	Decision
Airtel	ARIMA	0.2754	Adequate
	Holt	0.04614	Not Adequate
	ETS	0.05436	Adequate
BSNL	ARIMA	0.9455	Adequate
	Holt	0.751	Adequate
	ETS	0.9344	Adequate
Jio	ARIMA	0.8617	Adequate
	Holt	0.6772	Adequate
	ETS	0.06001	Adequate
VI	ARIMA	0.9992	Adequate
	Holt	0.7037	Adequate
	ETS	0.7405	Adequate

Table 3.5: Ljung–Box Test Results for Residuals of Telecom Models in India

Except Holt's method for Airtel, all models across operators pass the Ljung–Box test, indicating that residuals behave like white noise and the models are statistically adequate.

3.1.10 Model accuracy

To evaluate and compare the forecasting performance of the selected models, the accuracy measures Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) are computed on the test dataset. These measures help in quantifying the magnitude of prediction errors, where lower values indicate better model performance. The comparison of RMSE and MAE for ARIMA, Holt's Linear Trend, and ETS models across different operators is presented in the following table.

Operator	Model	RMSE	MAE
Airtel	ARIMA	3831583	1720269
	Holt	3930017	1833423
	ETS	3944755	1816559
BSNL	ARIMA	743222.5	521950.4
	Holt	770662	55479.5
	ETS	738424	519492.0
Jio	ARIMA	3744502	2304922
	Holt	7495151	4781521
	ETS	6386062	4608636
VI	ARIMA	9659884	4291927
	Holt	10123281	3915895
	ETS	10041134	3873789

Table 3.6: Comparison of RMSE and MAE for Different Models of Telecom Operators in India

It is observed that the forecasting accuracy varies across different operators and models. The ARIMA model performs better for Airtel and Jio as it provides comparatively lower RMSE and MAE values. For BSNL, Jio and VI the ETS model gives the minimum error values, indicating better performance.

operators	Best model
airtel	ARIMA(0,1,1) with drif
bsnl	ETS(A,Ad,N)
jio	ARIMA(0,2,1)(1,0,0)[12]
vi	ETS(A,Ad,N)

Table 3.7: Selected Best-Fit Models for Telecom Operators in India

3.1.11 Forecast Plot

Based on the best-fitted models identified and validated using accuracy measures, forecasts are generated for the future time periods.

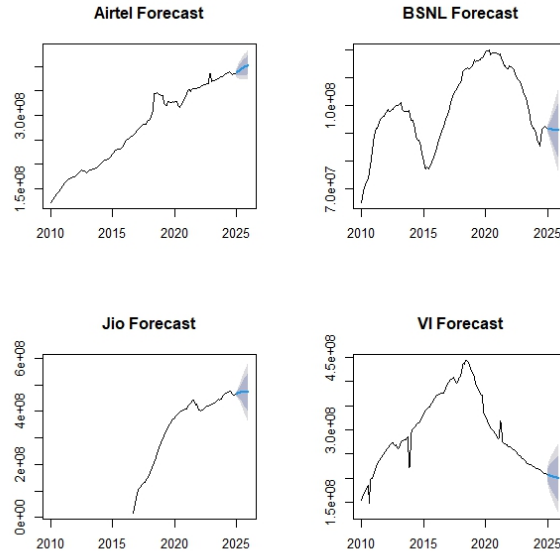


Figure 3.8: Forecast Plot of Telecom Subscribers in India Using Selected Models

The forecast analysis reveals distinct patterns among the four telecom operators. Airtel and Jio demonstrate stable and positive growth trends, indicating strong and consistent market positions with reliable future performance. In contrast, BSNL exhibits high volatility in its historical data, leading to a largely stagnant and uncertain future outlook with no clear growth direction. Similarly, VI continues to show a declining trajectory accompanied by high uncertainty, reflecting persistent weakness in its subscriber base. Overall, Airtel and Jio emerge as the market leaders with stable and predictable forecasting behavior, whereas BSNL and VI display weaker performance and greater uncertainty in their future trends.

3.2. Forecast values

To evaluate the forecasting performance more effectively, a comparison between actual and forecasted values was carried out. The forecast accuracy was assessed by computing the difference between actual and predicted values for each telecom operator. This helps in understanding how close the model predictions are to the real observed data.

operators	forecasted value 2025 march	actual 2025	Difference
airtel	389738451	389804037	65586
bsnl	91467180	91060664	406516
jio	469132261	469763852	631591
vi	205098822	205358733	259911

Table 3.8: Forecasted Telecom Subscriber Values for 2025 (India)

The above table presents a comparison between the forecasted values and the actual observed values for March 2025 for all telecom operators.

It can be observed that in some cases the forecasted values are slightly higher than the actual values, while in other cases the forecasted values are slightly lower than the actual values. This indicates that the models are capturing the overall trend but still exhibit small underestimation or overestimation depending on the operator and model behavior.

For Airtel, the forecasted value is slightly lower than the actual value, indicating a minor underestimation by the model. In the case of BSNL, the forecasted value is slightly higher than the actual value, showing a small overestimation. For Jio, the forecasted value is lower than the actual value, indicating under-prediction of the actual subscriber level. Similarly, for Vodafone Idea (VI), the forecasted value is slightly lower than the actual observed value, suggesting that the model underestimates the actual data.

The differences between actual and forecasted values are relatively small compared to the magnitude of the data, indicating that the models are performing well in capturing the underlying pattern. However, the presence of both positive and negative deviations suggests that the models are not perfectly unbiased and may slightly overestimate or underestimate depending on short-term fluctuations in the data.

Overall, the comparison shows that the forecasting models are reasonably accurate, as the deviations between actual and predicted values are minimal. This confirms that the selected models are suitable for short-term forecasting, although small prediction errors are still present due to random variations and external market influences.

Hence, the forecasting models provide a good approximation of actual values, with both overestimation and underestimation errors remaining within acceptable limits.

3.3. ANALYSIS OF OPARATERS IN KERALA

The following figure illustrates the time series trend of telecom subscribers for major operators in Kerala over the study period. The data captures the growth patterns and variations in subscriber base for Airtel, Jio, BSNL, and Vodafone Idea (VI). This visualization helps in understanding the comparative performance of each operator and identifying structural changes in the telecom market over time.

3.3.1 Time series plot

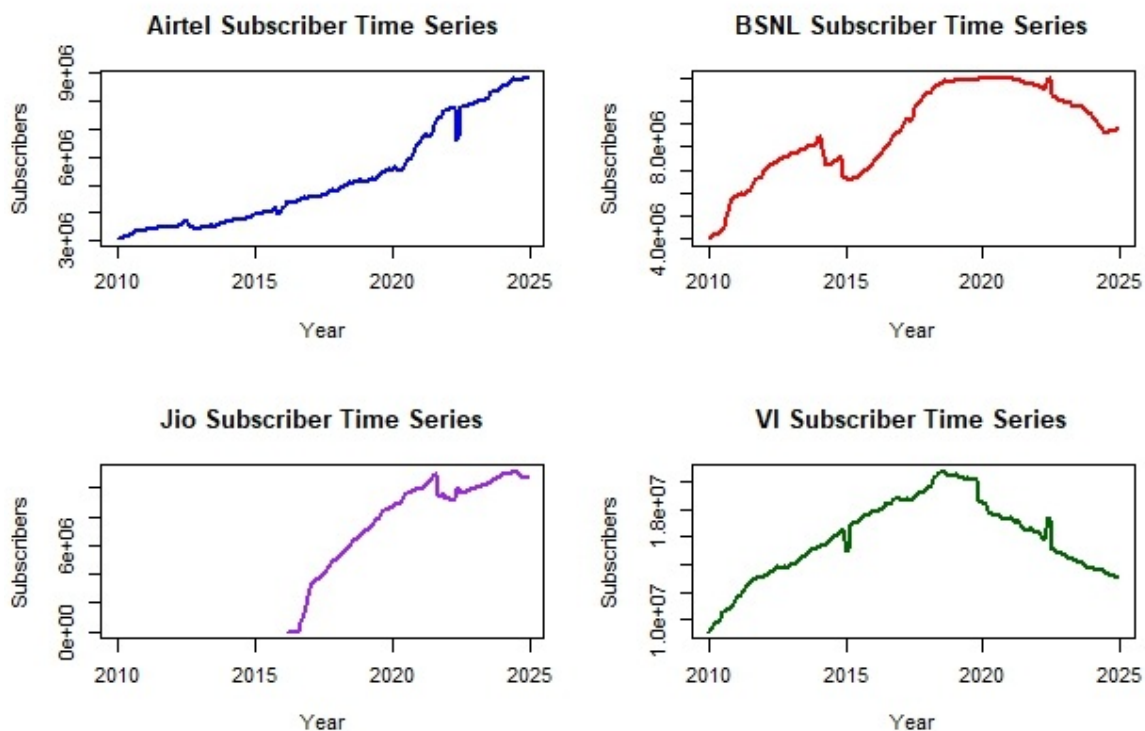


Figure 3.9: Time series plot of oparaters in Kerala

From the figure, it is clear that the telecom market in Kerala has undergone significant changes over the past 15 years. Airtel shows a steady upward trend with noticeable growth after 2020, indicating strong market expansion. Jio, entering around 2016, exhibits rapid growth and reaches a high subscriber base within a short period, though it begins to stabilize after 2022. In contrast, BSNL shows fluctuations with a peak around 2019–2020 followed by a decline. Vodafone Idea (VI), despite having the highest subscriber base initially, experiences a sharp decline after 2019. Overall, the figure reflects a shift in market dominance towards Airtel and Jio, while BSNL and VI lose subscribers.

3.3.2 Decomposition

The time series is decomposed into trend, seasonal, and irregular components to better understand the pattern in the data and to support appropriate model selection.

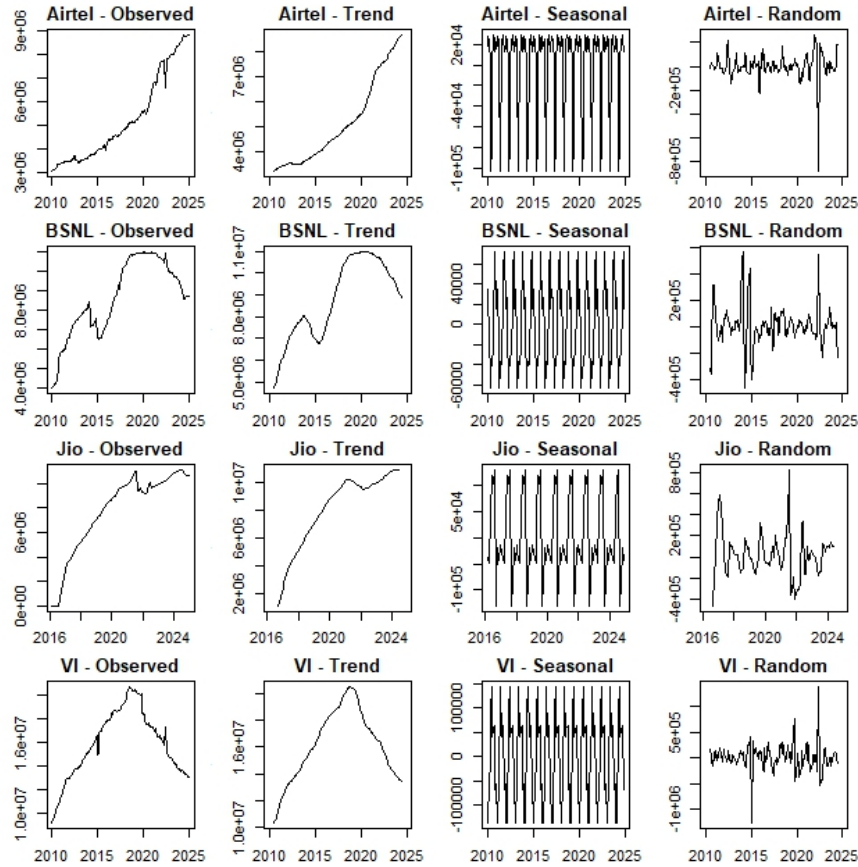


Figure 3.10: Decomposition Plot of operators in Kerala

For Airtel, there is a steady increase in subscribers. The trend is upward, seasonal variation is regular, and random fluctuations are small.

For BSNL, subscribers increase at first but later decrease. The trend shows a decline in recent years, and random variation is high.

For Jio, there is rapid growth after entry, followed by slight stabilization. The trend is strong, seasonal pattern is regular, and random variation is moderate.

For VI, subscribers increase initially and then decrease continuously. The trend shows a downward pattern in recent years, with noticeable random fluctuations.

Overall, all operators show clear trend and seasonal patterns, while randomness is higher for BSNL and VI.

3.3.3 Stationarity check

The stationarity of the subscriber time series data for Airtel, BSNL, Jio, and VI was examined using the Augmented Dickey-Fuller (ADF) test and KPSS test. The ADF test checks for the presence of a unit root, while the KPSS test verifies stationarity around a deterministic trend. The original time series of all operators were found to be non-stationary.

operators	adf value	kpss value	interpretaion
airtel	0.8616	0.01	Non-stationary
bsnl	0.9347	0.01	Non-stationary
jio	0.7075	0.01	Non-stationary
vi	0.963	0.01	Non-stationary

Table 3.9: Results of ADF and KPSS Tests for Stationarity of Telecom Operators in Kerala

3.3.4 Defferencing

Differencing was applied to transform the series into stationary form for further analysis and modeling.

operators	adf value	kpss value	interpretaion
airtel	0.01	0.07816	Stationary
bsnl	0.01	0.1	Stationary
jio	0.01	0.1	Stationary
vi	0.01	0.1	Stationary

Table 3.10: ADF and KPSS Test Results After First Differencing(Kerala)

The table presents the results of the ADF and KPSS tests after first differencing for all telecom operators in Kerala. The ADF test indicates that the null hypothesis of non-stationarity is rejected, while the KPSS test suggests that the null hypothesis of stationarity cannot be rejected. Since both tests consistently support stationarity

3.3.5 ACF and PACF

ACF and PACF plots are help in identifying the appropriate AR and MA terms for model selection.

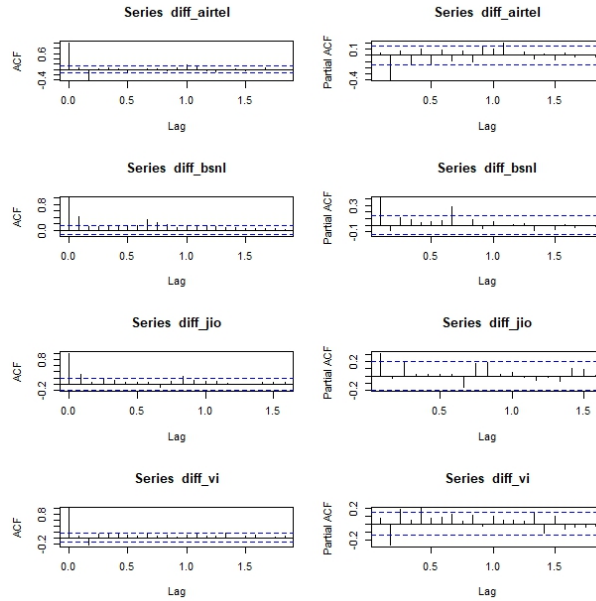


Figure 3.11: ACF and PACF Plots for Stationary Time Series (Kerala Telecom Operators)

Airtel shows a clear ACF cut-off at lag 1 with a gradually decaying PACF, indicating short-term dependence in the series. BSNL exhibits multiple significant spikes in both ACF and PACF, suggesting a more complex dependence structure. Jio shows a single significant spike in ACF at lag 1 with a decaying PACF, indicating limited short-term correlation. VI displays significant correlations at early lags in both ACF and PACF, indicating the presence of both autoregressive and moving-average type dependencies. Overall, Airtel and Jio show relatively simpler patterns, while BSNL and VI exhibit more complex dependency structures.

3.3.6 Model fitting

To forecast the subscriber data, three time series models—ARIMA, Holt’s method, and ETS—were applied for each operator. These models were chosen as they capture different patterns such as autocorrelation and trend.

Operator	ARIMA Model	Holt Model	ETS Model
Airtel	ARIMA(0,1,3)(0,0,1)[12] with drift	Holt (Linear Trend)	ETS(M,A,N)
BSNL	ARIMA(0,2,2)(1,0,0)[12]	Holt (Linear Trend)	ETS(A,Ad,N)
Jio	ARIMA(0,2,2)	Holt (Linear Trend)	ETS(M,N,A)
VI	ARIMA(2,2,1)	Holt (Linear Trend)	ETS(M,Ad,N)

Table 3.11: Different models of operators in kerala

3.3.7 Residual analysis

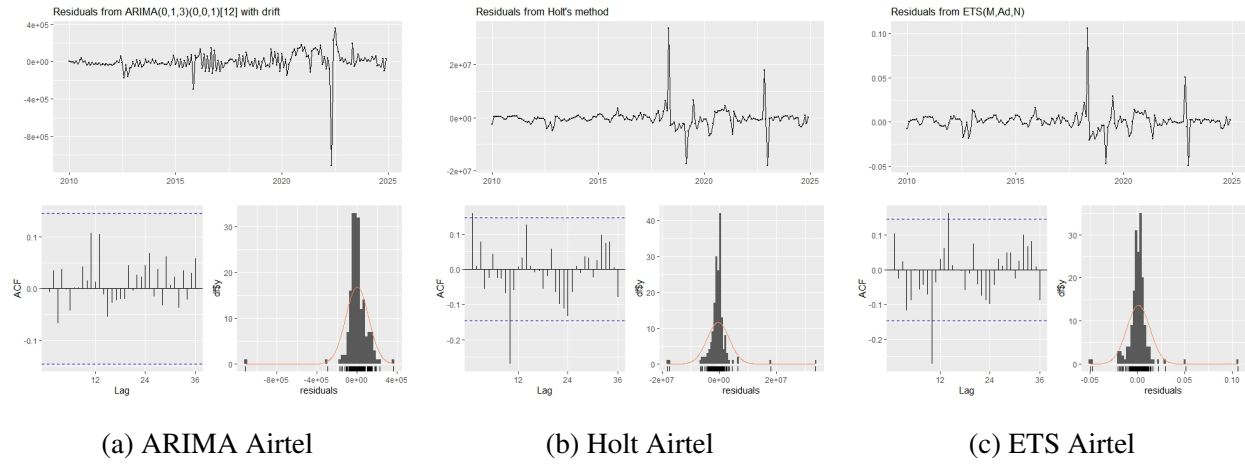


Figure 3.12: Forecasting models for Airtel in Kerala

The residual plots show that most values fluctuate randomly around zero with no clear trend. The ACF plots indicate that most autocorrelations lie within the confidence limits, with only a few minor spikes. The histogram appears approximately bell-shaped, suggesting near normality. However, a few large spikes are observed in the residual series, indicating occasional deviations. Overall, the residuals behave approximately like white noise with slight irregularities.

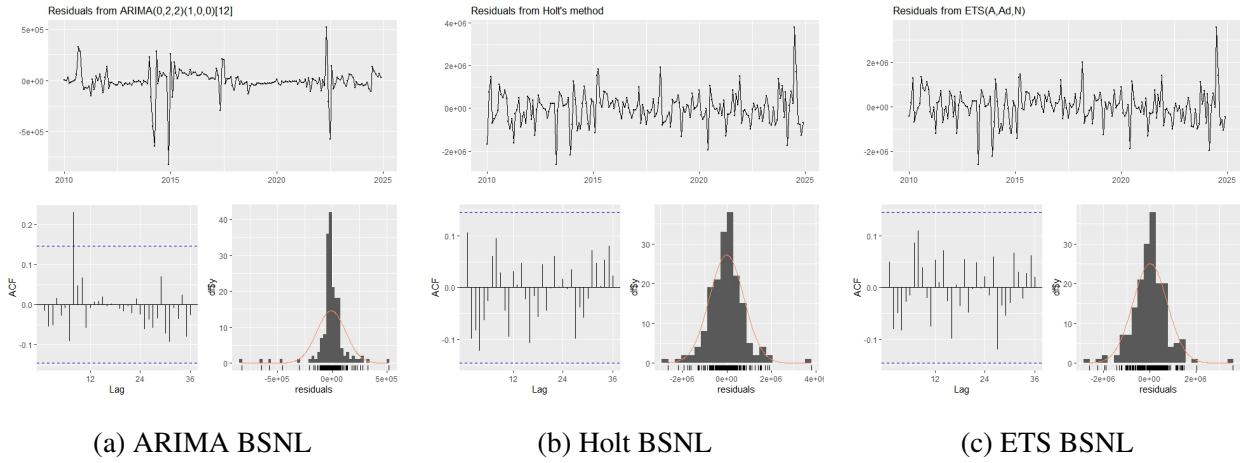


Figure 3.13: Forecasting models for BSNL in Kerala

The residuals appear randomly distributed around the mean without any visible trend. The ACF plots show that most values remain within the confidence limits, although a few spikes are present. The histogram is roughly bell-shaped, indicating approximate normality of residuals. Some sudden fluctuations are observed in the residual series, suggesting minor unexplained variations. Overall, the residuals are close to white noise with small deviations.

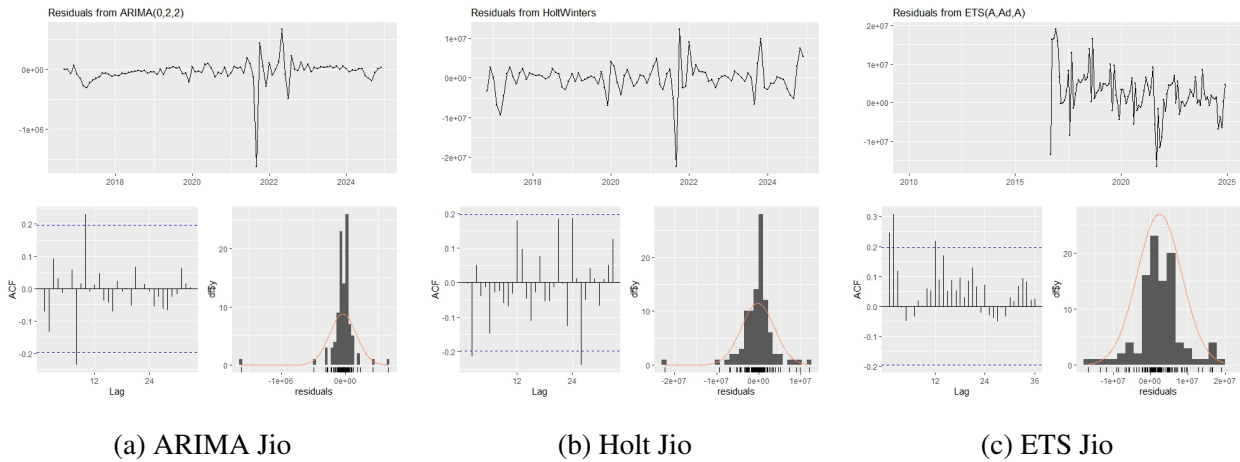


Figure 3.14: Forecasting models for VI in Kerala

The residual plots indicate random fluctuations around zero with no systematic pattern. The ACF plots show that most autocorrelation values lie within the confidence bounds, though a few spikes are visible. The histogram follows an approximately bell-shaped pattern, indicating near normal distribution. Some noticeable spikes in the residual series suggest occasional irregular behavior. Overall, the residuals are approximately white noise with slight departures.

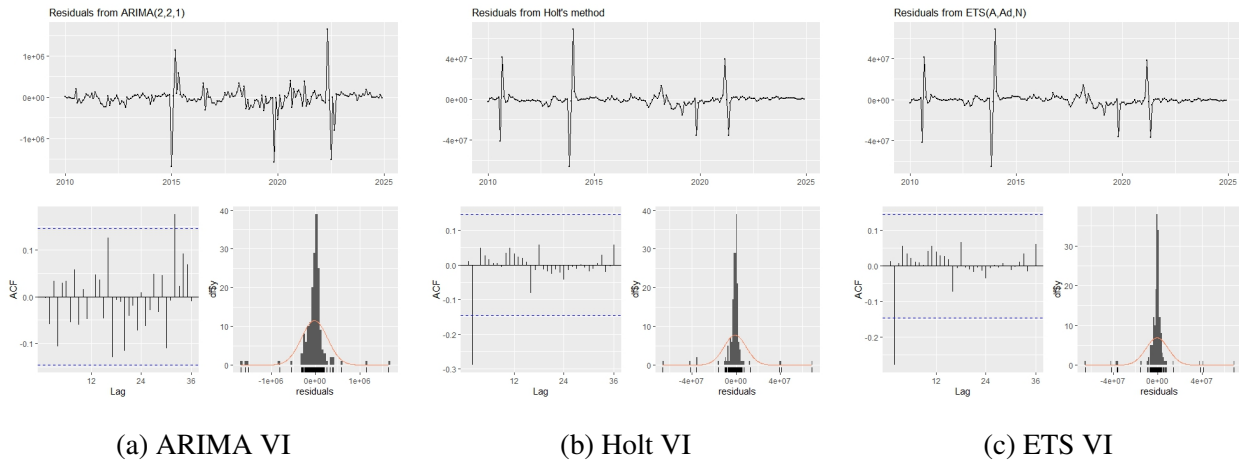


Figure 3.15: Forecasting models for VI in Kerala

The residuals are mostly centered around zero and do not exhibit any clear trend. The ACF plots indicate that most values fall within the confidence limits, with a few significant spikes. The histogram shows an approximately bell-shaped distribution, suggesting near normality. Some large fluctuations are present in the residual series, indicating occasional deviations. Overall, the residuals behave close to white noise with minor inconsistencies.

3.3.8 Ljung–Box test

Operator	Model	p value
Airtel	ARIMA	0.9865
	Holt	0.04614
	ETS	0.1055
BSNL	ARIMA	0.7949
	Holt	0.751
	ETS	0.07024
Jio	ARIMA	0.4703
	Holt	0.6772
	ETS	0.003586
VI	ARIMA	0.6668
	Holt	0.7037
	ETS	0.03552

Table 3.12: Ljung–Box Test Results for Residuals of Telecom Models in Kerala

The above table presents the Ljung-Box test p-values for residuals of different models across telecom operators. A p-value greater than 0.05 indicates that the residuals behave like white noise, implying that the model has adequately captured the underlying patterns in the data. It can be observed that ARIMA models generally produce higher p-values, indicating better residual

behavior compared to Holt and ETS models. In contrast, some ETS and Holt models show p-values less than 0.05, suggesting the presence of autocorrelation in residuals and inadequate model fit.

3.3.9 Model Accuracy

Operator	Model	RMSE	MAE
Airtel	ARIMA	113607.8	60968.04
	Holt	3930017	1833423
	ETS	133514.8	51375.98
BSNL	ARIMA	130081.5	70294.16
	Holt	770662	554379.5
	ETS	135759.4	70592.73
Jio	ARIMA	209910.8	101694.2
	Holt	556146.2	350014.1
	ETS	326092.9	235102.7
VI	ARIMA	294306.4	138402.7
	Holt	10123281	3915895
	ETS	316164.9	128335.8

Table 3.13: Comparison of RMSE and MAE for Different Models in Kerala

Based on the comparison of RMSE, MAE, and residual diagnostics, the best forecasting model for each telecom operator is selected. The selection is made by considering lower error values along with the adequacy of residual behavior. The following table presents the most suitable model for each operator.

operators	Best model
airtel	ARIMA(0,1,3)(0,0,1)[12] with drift
bsnl	ARIMA(0,2,2)(1,0,0)[12]
jio	ARIMA(0,2,2)
vi	ARIMA(2,2,1)

Table 3.14: Selected Best-Fit Models for Telecom Operators in Kerala

3.3.10 Forecast Plot

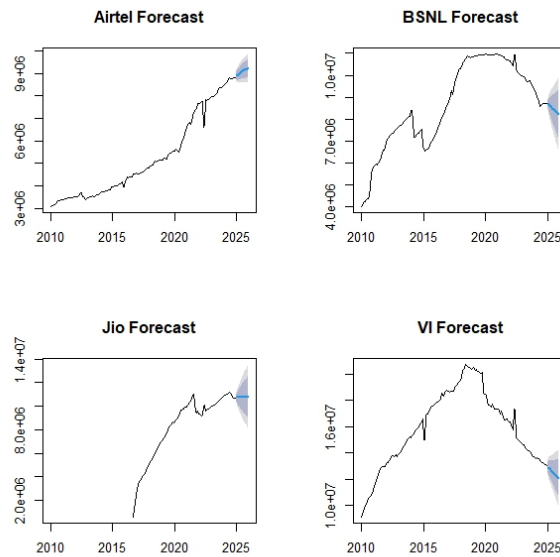


Figure 3.16: Forecast Plot of Telecom Subscribers in Kerala Using Selected Models

The figure presents the forecasted trends for Airtel, BSNL, Jio, and Vodafone Idea (VI) based on historical data. Airtel shows a steady and consistent upward trend, and the forecast indicates continued growth with relatively narrow confidence intervals, suggesting high reliability. BSNL exhibits considerable fluctuations with a declining trend in recent years, and the forecast suggests slight stabilization with higher uncertainty. Jio demonstrates rapid initial growth followed by a more stable phase, and the forecast indicates moderate future growth with some variability. In contrast, VI shows a clear downward trend, and the forecast predicts a continued decline with wide confidence intervals, indicating high uncertainty. Overall, Airtel and Jio display more stable and positive outlooks, whereas BSNL and VI show weaker and less predictable future performance.

3.3.11 Forecast Values

operators	forecasted value 2025 march	actual 2025	Difference
airtel	8955720	8892846	62874
bsnl	8603959	8780341	176382
jio	10742552	10699305	43247
vi	12787348	12920924	133576

Table 3.15: Comparison of Forecasted and Actual Subscriber Values for Kerala (2025)

The table indicates that the forecasted values are fairly close to the actual subscriber values for March 2025, demonstrating overall good model performance. Airtel and Jio show relatively smaller differences between forecasted and actual values, suggesting higher prediction accuracy and a better model fit. In contrast, BSNL and VI exhibit comparatively larger deviations, indicating slightly lower forecasting accuracy, possibly due to higher variability in their data. It is also important to note that for VI, the data used in the analysis corresponds to the period before the merger of Idea and Vodafone, which may have influenced the forecasting results. Despite these differences, the overall errors remain moderate, suggesting that the models are reasonably effective in capturing the underlying trends and patterns across all operators.

3.4. A STUDY OF FORECAST ACCURACY: COMBINING IDEA AND VODAFONE TO EVALUATE VI

The main objective of this study is to forecast the number of mobile subscribers for Idea and Vodafone separately for the period immediately following their merger, using appropriate time series models. These forecasts are developed to understand the expected behavior and trend of subscriber counts for both companies based on historical data. Time series modeling helps in capturing underlying patterns such as trend and short-term fluctuations, which are then used to project future values.

After obtaining the forecasted subscriber values for Idea and Vodafone, the study further aims to compare the combined forecasted values with the actual number of subscribers of Vodafone Idea (VI). This comparison is important because VI is formed through the merger of Idea and Vodafone. Therefore, if the subscribers of both companies are fully retained after the merger, the total number of VI subscribers should be approximately equal to the sum of the forecasted subscribers of Idea and Vodafone.

This analysis helps in understanding the consistency between pre-merger trends and post-merger performance, and also provides insight into the impact of the merger on the overall subscriber base.

3.4.1 Time series plot

The graphs show clear changes in subscriber levels, indicating the presence of a trend.

These plots help in selecting suitable forecasting models and form the basis for predicting subscriber values up to December 2020.

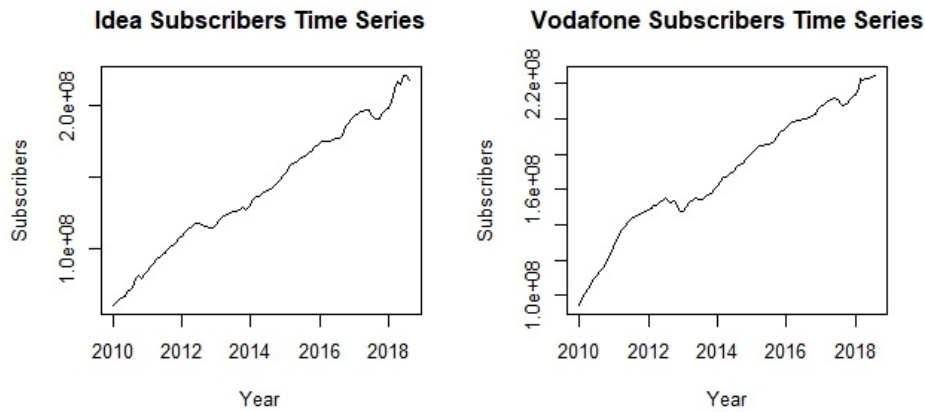


Figure 3.17: Time Series Plot of Idea and Vodafone Subscriber Data (India)

The plots present the historical trends of Idea and Vodafone before their merger in 2018. Idea shows a steady and consistent upward trend with slight cyclical fluctuations, indicating gradual growth with minor seasonal variations. Toward the end of the period, increased fluctuations suggest rising market instability. Vodafone also exhibits an overall upward trend but with distinct phases, including rapid growth, a short period of stagnation, and a later surge before 2018. Both companies demonstrate strong growth prior to the merger, which later combines into the more volatile pattern observed in VI.

3.4.2 Decomposition

The series is decomposed to separate the trend, seasonal, and irregular components for better understanding and modeling.

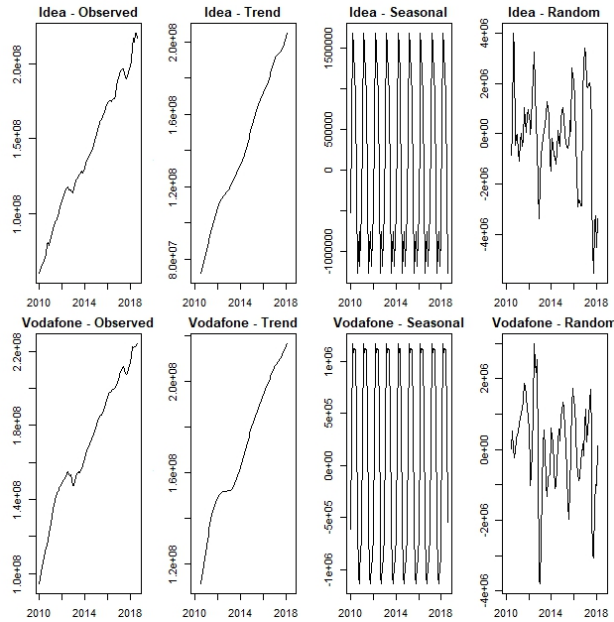


Figure 3.18: Decomposition of Idea and Vodafone Time Series Data

The time series was decomposed into trend, seasonal, and irregular components. The trend component shows the overall direction of the data, while the seasonal component captures repeating patterns over time.

The decomposition helps in better understanding the structure of the series and supports the selection of appropriate forecasting models.

3.4.3 Stationarity check

The stationarity of the subscriber time series data for Idea and Vodafone was examined using the Augmented Dickey-Fuller (ADF) Test and KPSS Test. The ADF test checks for the presence of a unit root, while the KPSS test verifies stationarity around a deterministic trend.

operators	adf value	kpss value	interpretaion
idea	0.01809	0.1	Stationary
vodafone	0.09843	0.01	Non-stationary

Table 3.16: Results of ADF and KPSS Tests for Stationarity of Idea and Vodafone

3.4.4 Differencing

Differencing is applied to achieve stationarity.

operators	adf value	kpss value	interpretaion
vodafone	0.09843	0.01	Non-stationary

Table 3.17: Results of ADF and KPSS Tests After Differencing for Idea and Vodafone

The results indicate that the Vodafone series is non-stationary, as the ADF test fails to reject the null hypothesis while the KPSS test supports non-stationarity. Hence, differencing is required to make the series stationary.

3.4.5 Second Differencing

operators	adf value	kpss value	interpretaion
vodafone	0.01	0.1	Stationary

Table 3.18: Results of ADF and KPSS Tests After Second Differencing for Idea and Vodafone

3.4.6 ACF and PACF

The stationary series is further analyzed using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots to determine the suitable orders of the AR and MA components for model identification.

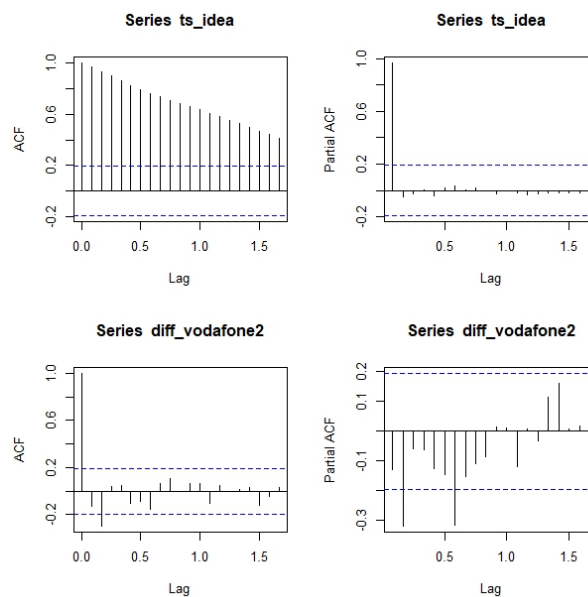


Figure 3.19: ACF and PACF Plots for Stationary Series of Idea and Vodafone

The ACF and PACF plots for Idea show a slow decay pattern and a strong spike at lag 1, indicating non-stationarity due to a strong trend in the data. Therefore, differencing is required before model fitting. In contrast, the ACF and PACF plots for Vodafone after second differencing show most values within the confidence limits, indicating that the series has become stationary. The presence of significant spikes at early lags suggests that a moving average component may be suitable for modeling.

3.4.7 Model fitting

operators	MODEL
idea	ARIMA(0,1,1)(1,0,0)[12] with drift
idea	ETS(M,A,N)
idea	Holt model
vodafone	ARIMA(0,2,2)(1,0,0)[12]
vodafone	ETS(M,Ad,N)
vodafone	Hlot model

Table 3.19: Selected Models for Idea and Vodafone Time Series Data

3.5. Residual Analysis

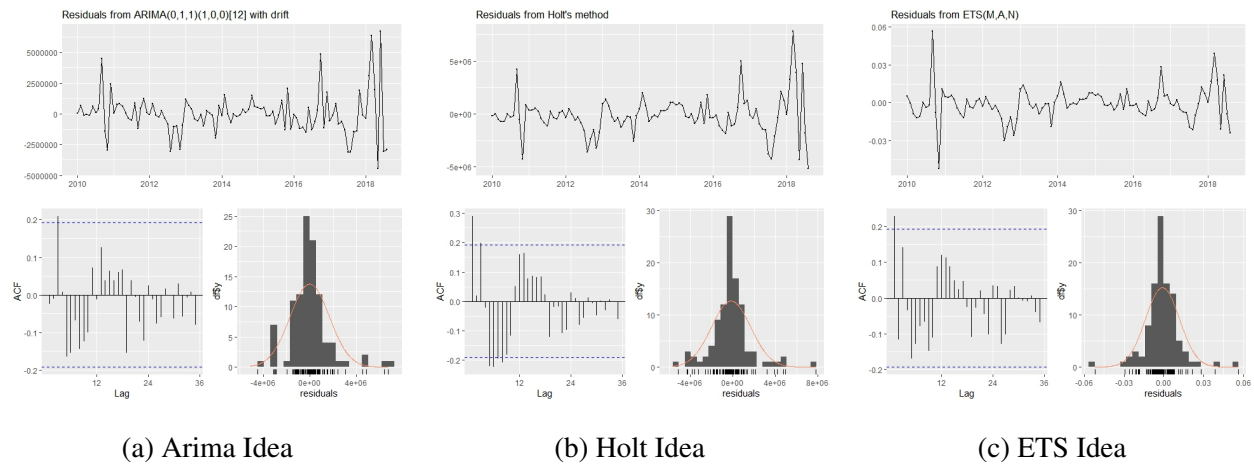


Figure 3.20: Residual Plots of Idea

The ARIMA model shows residuals centered around zero with most ACF values within the confidence limits, indicating that it effectively captures both trend and seasonal patterns. In contrast, Holt's and ETS models exhibit significant spikes at lag 12 in the ACF plots, suggesting that seasonal effects are not properly modeled. Additionally, these models show relatively higher variability in residuals. Overall, ARIMA performs better than the other models, as it produces more reliable and near white-noise residuals, making it the most suitable model for Idea data.

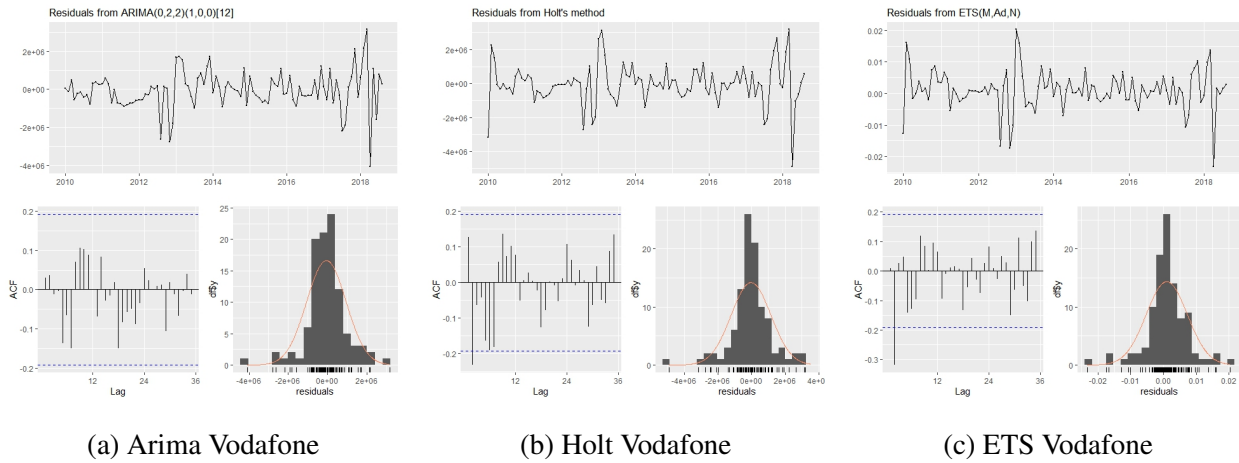


Figure 3.21: Residual Plots of Vodafone

The ARIMA model shows residuals centered around zero with most ACF values within the confidence limits, indicating that it successfully captures both trend and seasonal patterns. In contrast, Holt's and ETS models display significant spikes at seasonal lags (such as lag 12), suggesting that seasonality is not properly accounted for. These models also exhibit higher variability in residuals. Overall, the ARIMA model performs better, as it produces more stable and near white-noise residuals, making it the most suitable model for Vodafone data.

3.5.1 Ljung–Box test

Operator	Model	p value
Idea	ARIMA	0.1838
	Holt	0.000218
	ETS	0.166
Vodafone	Holt	0.6273
	ETS	0.1505

Table 3.20: Ljung–Box Test Results for Idea and Vodafone Models

The Ljung–Box test results indicate that most models produce residuals that do not exhibit significant autocorrelation, implying an adequate fit to the data. However, Holt's method for Idea shows a very low p-value, suggesting the presence of autocorrelation and hence a poor model fit. For Vodafone, all models yield acceptable p-values, indicating that they adequately capture the underlying patterns. Overall, ARIMA provides the most consistent and reliable performance across both datasets.

3.5.2 Model Accuracy

To evaluate the forecasting accuracy of the fitted models, performance measures such as RMSE and MSE were computed. These measures help in assessing the prediction error of each model. Lower values of RMSE and MSE indicate better forecasting performance. The results obtained for the ARIMA, Holt, and ETS models are presented in the following table.

operators	MODEL	RMSE	MSE
	ARIMA(0,1,1)(1,0,0)[12] with drift	1679839	1080944
idea	Holt	1845812	1212560
	ETS(M,A,N)	1849030	1207611
	ARIMA(0,2,2)(1,0,0)[12]	1002655	707616.7
vodafone	Holt	1177574	780613.7
	ETS(M,Ad,N)	1085494	720816.6

Table 3.21: Comparison of Model Accuracy for Idea and Vodafone

The model accuracy results based on RMSE and MSE indicate that the ARIMA model provides the best performance for both Idea and Vodafone, as it yields the lowest error measures. In contrast, Holt and ETS models show comparatively higher errors, suggesting less accurate forecasts. Although ETS performs slightly better than Holt in some cases, it still does not outperform ARIMA. Overall, ARIMA is the most suitable model for both datasets due to its superior predictive accuracy.

3.5.3 Forecast

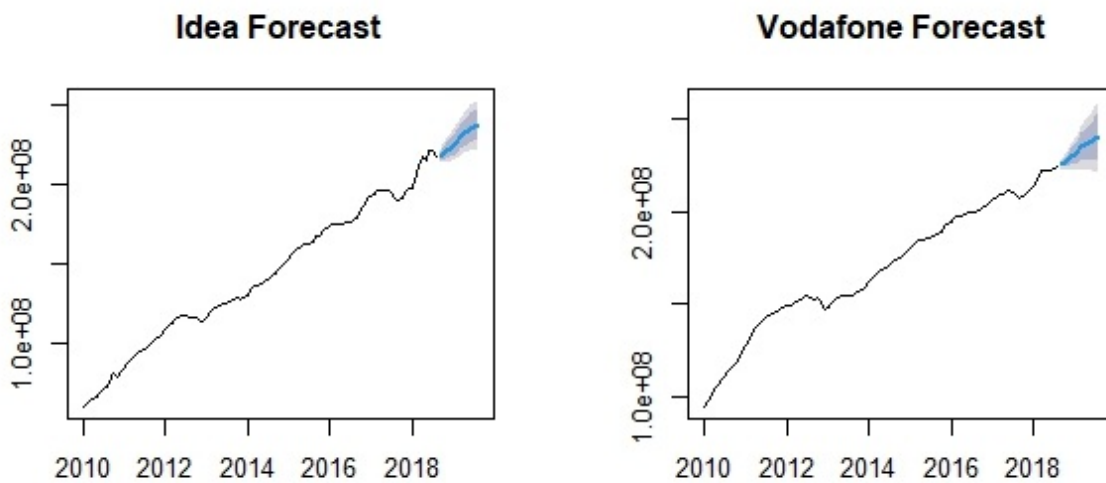


Figure 3.22: Forecast Plot for Idea and Vodafone Subscriber Data

The forecast plots for Idea and Vodafone show a continued upward trend based on historical data. For Idea, the forecast indicates steady growth with relatively narrow confidence intervals, suggesting higher reliability of predictions. Similarly, Vodafone shows an increasing trend, though with slightly wider confidence intervals, indicating more uncertainty.

3.5.4 Forecast Values

operators	Forcaste value September 2018
idea	217141707
vodafone	225525280

Table 3.22: Forecasted Subscriber Values of Idea and Vodafone Just After Merger

These values represent the projected performance just before the merger period, reflecting the growth trend observed in the historical data. The forecasts suggest that both series maintained an increasing pattern up to this point, consistent with their overall upward trajectory prior to the structural changes in the telecom market.

The forecasted subscriber values of Idea and Vodafone were summed and compared with the actual VI (Vodafone Idea) subscriber data. This helps in understanding whether the total forecasted subscribers align with the actual combined subscriber base. Any differences between these values indicate changes such as user migration or model inaccuracies.

Month	idea+vodafone	vi actual subscribers	difference
September 2018	442666987	434964406	7702581

Table 3.23: Comparison of Combined Forecasted Subscribers (Idea + Vodafone) and Actual VI Subscribers with Difference

The difference between the combined forecasted subscribers of Idea and Vodafone and the actual VI subscribers indicates that the merger did not fully achieve the expected outcome. Since the actual value is lower than the forecasted value, it suggests a slight underperformance after the merger. This may be due to factors such as integration challenges, network issues, increased competition (especially from Jio and Airtel), customer churn, and pricing pressures in the telecom market. Therefore, the merger was not completely satisfactory in achieving the expected subscriber base.

Chapter 4

Conclusion

This study analyzed the subscriber trends of major telecom operators using time series techniques and provided forecasts to understand future market behavior. The analysis was carried out at both the national (India) and regional (Kerala) levels, along with a focused study on the merger of Idea and Vodafone to evaluate the performance of Vodafone Idea (VI).

From the analysis of telecom operators in India, it is observed that Airtel and Jio exhibit strong and consistent growth patterns, indicating stable market positions and long-term sustainability. In contrast, BSNL and VI show declining or highly fluctuating trends, reflecting weaker competitiveness and challenges in retaining subscribers. The forecasting models, particularly ARIMA and ETS, performed effectively in capturing the underlying patterns, and the forecasted values were found to be very close to the actual observations, confirming good model accuracy.

A similar pattern is observed in the Kerala telecom market. Airtel and Jio continue to dominate with steady growth and relatively stable forecasts, while BSNL and VI demonstrate declining trends and higher uncertainty. Although regional variations exist, the overall market dynamics in Kerala closely follow the national pattern, indicating a consistent shift in subscriber preference toward private telecom operators.

The study also highlights the importance of model selection. ARIMA models generally provided better performance due to their ability to capture both trend and autocorrelation, while ETS models performed well in certain cases such as BSNL. Residual diagnostics and Ljung–Box tests confirmed that most selected models were statistically adequate, and the forecast errors (RMSE and MAE) remained low, indicating reliable short-term forecasting.

A key contribution of this study is the analysis of the Idea–Vodafone merger. The combined forecasted subscriber base of Idea and Vodafone was found to be higher than the actual subscriber

base of VI after the merger. This indicates that the merger did not fully retain the expected number of subscribers. The observed gap suggests the presence of subscriber loss, which may be attributed to factors such as increased competition, particularly from Jio and Airtel, customer migration, pricing strategies, and integration challenges during the merger process.

Overall, the study demonstrates that time series models are effective tools for analyzing telecom subscriber data and generating accurate forecasts. The results provide valuable insights into market trends, competitive dynamics, and the impact of structural changes such as mergers. The findings are useful for telecom companies in strategic planning and for policymakers in understanding market behavior and ensuring a competitive telecom environment.

In conclusion, Airtel and Jio emerge as strong and stable operators with positive future prospects, while BSNL and VI face challenges in maintaining their subscriber base. The study successfully integrates statistical modeling with real-world telecom data to provide meaningful insights into the evolving telecom sector.

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APPENDIX

R Code for Time Series Analysis and Forecasting

```
# Step 1: Load Required Libraries
library(readxl)
library(tseries)
library(forecast)
library(ggplot2)

# Step 2: Import Data
data <- read_excel(____, sheet = "____")
data$month <- match(tolower(data$month), tolower(month.abb))

# Step 3: Create Time Series Data
ts_ <- ts(data$, start = c(2010,1), frequency = 12)
# Jio (starts from 2016)
start_jio <- which(data$year == 2016 & data$month == 9)
jio_clean <- data$jio
jio_clean[1:(start_jio-1)] <- NA
ts_jio <- ts(jio_clean, start = c(2010,1), frequency = 12)

# Step 4: Plot Time Series
plot(ts_, main="", col="blue")

# Step 5: Stationarity Test (ADF)
adf.test(ts_)

# Differencing
diff_ <- diff(ts_)

# ADF after differencing
adf.test(diff_)

# Step 6: ACF and PACF Plots
acf(diff_); pacf(diff_)

# Step 7: Model Fitting (ARIMA)
model_ <- auto.arima(ts_)

# Model Summary
summary(model_)
```

```

# Step 8: Model Accuracy
accuracy(model_)

# Step 9: Residual Diagnostics
checkresiduals(model_)

# Step 10: Forecasting (ARIMA)
fc_ <- forecast(model_, h=3)

# Extract Forecast Values
fc_$mean[3]

# Step 11: Compare with Actual Values
actual_ <- ----
actual_oparater - fc_oparater$mean[3]

# Step 12: ETS Model
model_ets_a <- ets(ts_a)

# Forecast ETS
forecast(model_ets_, h=3)

# Step 13: Holt Model
model_Holt_ <- Holt(ts_)

# Step 14: Forecast Plots
plot(forecast(model, h=12), main=" Forecast")

# Step 15: Best Models
best_oparater <- model_oparater

# Residual check for best models
checkresiduals(best_oparater)

```